A SMOOTH MAP OF AFFINE SCHEMES DESCENDS TO A SMOOTH MAP OF FINITE TYPE Z-SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9827-9833 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9835-9842, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The ring-theoretic characterization of smooth morphisms of affines (Morphisms, smooth-characterize) and the algebra lemma approximating a smooth algebra of finite presentation over a smooth algebra over a Noetherian base (Algebra, finite-presentation-fs-Noetherian); cited inputs with the translation written in the retained span.

Source reference. Original lemma 00TP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9842–9842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05FB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9739–9753. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01V6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9842–9842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of affine schemes, which is smooth. Then there exists a diagram as in Lemma 05FB such that in addition f_0 is smooth.*

Proof. Write $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and as f is smooth we see that B is smooth as an A-algebra, see Morphisms, Lemma 01V6. Hence the lemma follows from Algebra, Lemma 00TP. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.